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[特別講演]FxLMSアルゴリズムを用いた 適応アクティブノイズコントロール – 理論から応用 –

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Adaptive Active Control of Acoustic Noise Using FxLMS Algorithm: from Theory to Practice

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Abstract This paper studies theoretical analysis and practical implementation of adaptive active noise control (ANC) systems using Filtered-x Least Mean Square (FxLMS) algorithm. A comprehensive theoretical model for the adaptation process performed by the FxLMS algorithm is developed. Based on this model, closed-form mathematical expressions for formulating behaviors of FxLMS-based ANC systems are derived. These expressions are derived without using any simplifying assumption regarding the acoustic plant and noise. Also, possible modeling errors are considered. Consequently, the obtained expressions can apply to a practical case with realistic working conditions. The validity of the theoretical results is verified by using a fully-implemented ANC system.

Key words FxLMS, Active Noise Control, Convergence Analysis,

1. Introduction

The traditional approach to acoustic noise control uses passive silencers to attenuate unwanted sound waves. These silencers are valued for their global noise attenuation; however, they are relatively bulky, costly and ineffective for low frequency noise [1]. To overcome these problems, *Active Noise Control* (ANC) has received considerable interest. The first patent on ANC was granted to Paul Leug, in 1936 [2]. Leug appended a drawing to his patent, describing how two sinusoidal sound waves can cancel each other. For two decades this idea remained only a theory on paper until Olson used early analog electronic technology to invent the first ANC device, called the "electronic sound absorber" [3], [4]. By the end of the 1950's, several analog ANC devices were invented, including those patented by Fogel [5], Simshauser [6] and Bose. However, all of them were not able to adapt to changing characteristics of the noise to be canceled nor to changing environmental conditions. This was because adaptive signal processing techniques could not be realized by using analog electronic technology. Only with the advent of digital technology did the realization of adaptive ANC systems become possible. The theory of adaptive ANC was established by Widrow in 1975 [7]; however the most significant progresses on this subject has been reported in the pas decade.

Fig. 1 shows a general block diagram for adaptive ANC with single-channel feed-forward structure [8], [9]. $x(n)$, $y(n)$ and $e(n)$ represent reference noise, anti-noise and residual noise in the electric domain, respectively. $x(n)$ is obtained by measuring the sound pressure at a location close to the noise source. $e(n)$ is obtained by measuring the sound pressure at the desired zone of silence, Z_s. $y(n)$ is generated in the electric domain and fed to a loudspeaker. The sound generated by this loudspeaker propagates through an acoustic channel, S, to reach Z_s (secondary path). On the other hand, the noise generated by the noise source, propagates through another acoustic channel, P, to reach Z_s (primary path). These two signals combine acoustically with each other across the medium; however, $e(n)$ that is the net sound pressure at Z_s is only taken as a measure of the residual noise. Filtered-x Least-Mean-Square (FxLMS) algorithm adaptively minimize $e(n)$ in the acoustic domain. This algorithm computes $y(n)$ as the response of an adaptive filter, W, to $x(n)$ and, also, updates the weight vector of W, $\mathbf{w}(n)$.

There have been several contributions in the modeling and analysis of the FxLMS adaptation process [10]〜[18]. How-

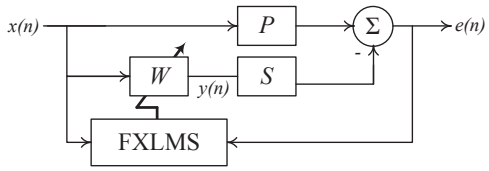


図 1　シングルチャネルフィードフォワード ANC のブロック図
Fig. 1　Block diagram for single-channel feed-forward ANC

ever, only a few have intended to find closed-form expressions to model this process. Even if such expressions were derived, simplified cases had to be considered. Long summarized early work on this subject [12] and derived closed-form expressions for the stability bound (μ_{max}) and steady state performance (J_{ss}) of this process. However, he assumed a pure delay secondary path. Elliott derived another expression for μ_{max} [14], which has become more popular than Long's expression. Bjarnason conducted a comprehensive analysis on the FxLMS adaptation process [15]. However, once he intended to derive closed-form expressions for μ_{max} and J_{ss}, he had to simplify his model by assuming a pure delay secondary path, a perfectly accurate secondary path model ($\hat{\mathbf{s}}$=$\mathbf{s}$), and a broad-band input signal. Also, Vicente derived another expression for μ_{max} when the acoustic noise is assumed to be sum of deterministic sinusoids [19].

All the aforementioned analysis were carried out for pure delay secondary paths, perfectly accurate secondary path models and broad-band noise. Accordingly, a practical μ_{max} is different with those have been theoretically proposed [20]. The authors have investigated behaviors of the FxLMS adaptation process in practical conditions. This investigation has led to a set of closed-form expressions for formulating behaviors of the FxLMS adaptation process. The results have been published in a series of papers [16]〜[18]. In [16], basic closed-form expressions for formulating behaviors of the FxLMS adaptation process with a general secondary path were derived. In [17], these expressions were generalized by considering the effects of the noise band-width. In [18], these expressions were generalized further by taking the influences of the imperfect secondary path models into account. The relative drawback of the analysis conducted in [17] is that it assumes a perfect secondary path model. The relative drawback of the analysis conducted in [16] is that it assumes a broad-band noise. This paper aims at removing all the relative drawbacks of the previous works, resulting in a more comprehensive analytical expressions for describing behaviors of FxLMS-based ANC.

2. Theoretical Modeling of FxLMS-based ANC

The FxLMS algorithm is given by

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu e(n) \hat{\mathbf{s}}^T \mathbf{x}(n) \tag{1}$$

where μ is the step-size, $\hat{\mathbf{s}}$ is a vector containing a model of the secondary path impulse response ($\mathbf{s}$), and $\mathbf{x}(n)$ is a $L \times 1$ tap vector of $x(n)$. The mean-square-error (MSE) of the FxLMS algorithm, $J(n)$ is expressed as [16]

$$J(n) = J_o + \sum_{q=0}^{Q-1} s_q^2 E\left\{\mathbf{c}^T(n-q)\,\mathbf{\Lambda}\mathbf{c}(n-q)\right\} \tag{2}$$

where J_o is the minimal MSE level, $\mathbf{\Lambda}$ is the diagonalized auto-correlation matrix of $x(n)$, s_q denotes elements of $\mathbf{s}$, Q is the length of $\mathbf{s}$ and $\mathbf{c}(n)$ is the deviation of $\mathbf{w}(n)$ from its optimal vector. The excess-MSE function, $J_{ex}(n)$ is defined as a dynamic measure, determining the deviation of $J(n)$ from its minimal level; thus,

$$J_{ex}(n) = \sum_{q=0}^{Q-1} s_q^2 E\left\{\mathbf{c}^T(n-q)\,\mathbf{\Lambda}\mathbf{c}(n-q)\right\} \tag{3}$$

In order to investigate the variation of $J_{ex}(n)$, its time difference can be calculated from Eq. (3) by using the same logic used in [18]:

$$\triangle J_{ex}(n) = J_{ex}(n+1) - J_{ex}(n) = \tag{4}$$

$$\mu^2 \lambda_{rms}^2 \|\hat{\mathbf{s}}\|^2 \|\mathbf{s}\|^2 L J_o - \mu \sum_{q,p=0}^{Q-1} \sum_{l=0}^{L-1} \gamma_{l,p,q} E\left\{c_l^2(n-p-q)\right\}$$

where

$$\gamma_{l,p,q} = \lambda_l s_q^2 \left(2\lambda_l s_p \hat{s}_p - \mu \lambda_{rms}^2 \|\hat{\mathbf{s}}\|^2 \Delta\right) \tag{5}$$

$$\Delta = L s_p^2 + 2\left(\frac{\lambda_l}{\lambda_{rms}}\right)^2 s_p \hat{s}_p \frac{\mathbf{s}^T \mathbf{\Psi} \hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|^2} \tag{6}$$

In Eqs. (4)-(6), λ_l shows Eigenvalues of $\mathbf{\Lambda}$, λ_{rms} is the root-mean-square of λ_l and $\mathbf{\Psi}$ is given by

$$\mathbf{\Psi} = diag(0, 1, \ldots, Q-1), \tag{7}$$

Eq. (4) can be simplified by assuming a broad-band white noise; however, a realistic noise can be only white over a limited frequency range (band-limited white). In [17], it is shown that for a band-limited white noise with band-width of B_w and power of σ_x^2, the Eigenvalues $\lambda_0, \ldots, \lambda_{L-1}$ can only be equal to zero or $\lambda_l = B_w^{-1}\sigma_x^2$. Also, it is shown that $\lambda_{rms} = B_w^{-0.5}\sigma_x^2$. In this case, $\gamma_{l,p,q} \neq 0$ only when $\lambda_l \neq 0$. Accordingly, Eqs. (5) and (6) can be re-expressed as

$$\gamma_{l,p,q} = \lambda_l \frac{\sigma_x^2}{B_w} s_q^2 \left(2 s_p \hat{s}_p - \mu \sigma_x^2 \|\hat{\mathbf{s}}\|^2 \Delta\right) \tag{8}$$

$$\Delta = L s_p^2 + \frac{2}{B_w} s_p \hat{s}_p \frac{\mathbf{s}^T \mathbf{\Psi} \hat{\mathbf{s}}}{\|\hat{\mathbf{s}}\|^2} \tag{9}$$

Substituting Eqs. (8) and (9) into Eq. (4) results in

$$\triangle J_{ex}(n) = \mu^2 \frac{\sigma_x^4}{B_w} \|\hat{\mathbf{s}}\|^2 \|\mathbf{s}\|^2 L J_o \tag{10}$$

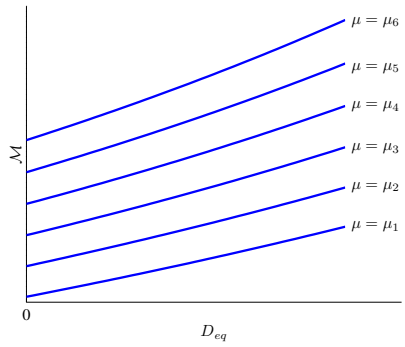
$$+\mu^2 \frac{\sigma_x^4}{B_w} \|\hat{\mathbf{s}}\|^2 L \sum_{p=0}^{Q-1} s_p^2 J_{ex}(n-p)$$

$$-2\mu \frac{\sigma_x^2}{B_w} \left(1 - \mu \frac{\sigma_x^2}{B_w} \mathbf{s}^T \mathbf{\Psi} \hat{\mathbf{s}}\right) \sum_{p=0}^{Q-1} s_p \hat{s}_p J_{ex}(n-p)$$

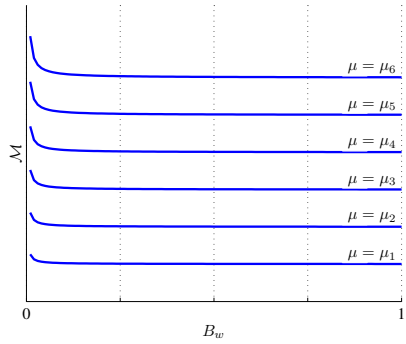
For a slow adaptation process, it can be shown that

$$\sum_{p=0}^{Q-1} s_p \hat{s}_p J_{ex}(n-p) = \frac{\mathbf{s}^T \hat{\mathbf{s}}}{\|\mathbf{s}\|^2} \sum_{p=0}^{Q-1} s_p^2 J_{ex}(n-p) \tag{11}$$

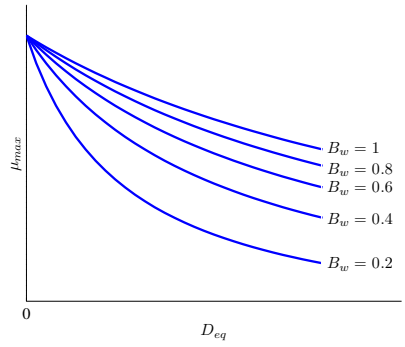
Substituting Eq. (11) into (10) results in



(a)



(b)



(c)

図 2　ステップサイズが $\mathcal{M}$ における D_{eq} と B_w ならびに μ_{max} に与える影響 ($\mu_1<\mu_2<\mu_3<\mu_4<\mu_5<\mu_6$)

Fig. 2　Influences of D_{eq} and B_w on $\mathcal{M}$ and μ_{max} for different values of step-size ($\mu_1<\mu_2<\mu_3<\mu_4<\mu_5<\mu_6$)

$$\triangle J_{ex}(n)=\mu^2\frac{\sigma_x^4}{B_w}\left\|\hat{\mathbf{s}}\right\|^2\left\|\mathbf{s}\right\|^2 LJ_o+\left[\mu^2\frac{\sigma_x^4}{B_w}\left\|\hat{\mathbf{s}}\right\|^2 L-\right. \quad (12)$$

$$\left.2\mu\frac{\sigma_x^2}{B_w}\frac{\mathbf{s}^T\hat{\mathbf{s}}}{\left\|\mathbf{s}\right\|^2}\left(1-\mu\frac{\sigma_x^2}{B_w}\mathbf{s}^T\mathbf{\Psi}\hat{\mathbf{s}}\right)\right]\sum_{p=0}^{Q-1}s_p^2 J_{ex}(n-p)$$

The distinction between this model and the one derived in [18] is that this model considers the noise bandwidth.

2. 1　Steady-State Analysis

Assuming a stable adaptation process, it can be shown that $\triangle J_{ex}(n)$=0 and $J_{ex}(n+1)=J_{ex}(n)=J_{ex}(n-p)=J_{ex}(\infty)$ in steady state conditions. Thus, setting Eq. (12) to zero results in

$$J_{ex}(\infty)=\frac{\mu\left\|\mathbf{s}\right\|^2\sigma_x^2 LJ_o}{2\frac{\mathbf{s}^T\hat{\mathbf{s}}}{\left\|\hat{\mathbf{s}}\right\|^2}-\mu\left\|\mathbf{s}\right\|^2\sigma_x^2\left(L+\frac{2}{B_w}\frac{\mathbf{s}^T\hat{\mathbf{s}}\mathbf{s}^T\mathbf{\Psi}\hat{\mathbf{s}}}{\left\|\hat{\mathbf{s}}\right\|^2\left\|\mathbf{s}\right\|^2}\right)} \quad (13)$$

The steady-state performance is usually measured using a relative parameter, called the *misadjustment* $\mathcal{M}$, which is defined as the ratio of $\triangle J_{ex}(n)$ to J_o. The steady-state level of the MSE function can be then formulated as

$$J_{ss}=(1+\mathcal{M})J_o \quad (14)$$

From Eq. (13), $\mathcal{M}$ is obtained as

$$\mathcal{M}=\frac{\mu\left\|\mathbf{s}\right\|^2\sigma_x^2 L}{2\frac{\mathbf{s}^T\hat{\mathbf{s}}}{\left\|\hat{\mathbf{s}}\right\|^2}-\mu\left\|\mathbf{s}\right\|^2\sigma_x^2\left(L+\frac{2}{B_w}\frac{\mathbf{s}^T\hat{\mathbf{s}}\mathbf{s}^T\mathbf{\Psi}\hat{\mathbf{s}}}{\left\|\hat{\mathbf{s}}\right\|^2\left\|\mathbf{s}\right\|^2}\right)} \quad (15)$$

For a perfect secondary path model ($\hat{\mathbf{s}}$=$\mathbf{s}$), $\mathcal{M}$ is simplified to $\bar{\mathcal{M}}$:

$$\bar{\mathcal{M}}=\frac{\mu\left\|\mathbf{s}\right\|^2\sigma_x^2 L}{2-\mu\left\|\mathbf{s}\right\|^2\sigma_x^2\left(L+\frac{2}{B_w}D_{eq}\right)} \quad (16)$$

where the equivalent delay of the secondary path is defined as

$$D_{eq}=\frac{\mathbf{s}^T\mathbf{\Psi}\mathbf{s}}{\left\|\mathbf{s}\right\|^2}$$

As can be seen in Eq. (16), the influences of the secondary path on $\mathcal{M}$ is taken into account by two scalar parameters: $\|\mathbf{s}\|$, and D_{eq}. The influence of $\|\mathbf{s}\|$ can be compensated for by adjusting μ (normalizing by the filtered-reference signal power that is $\|\mathbf{s}\|^2\sigma_x^2$). Therefore, D_{eq} can be considered as the major parameter of the secondary path which affects on the steady-state performance of the system. By using Eq. (16), the variation of $\mathcal{M}$ with respect to D_{eq} can be plotted as shown in Figure 2-a. From this figure, it can be deduced that D_{eq} causes the steady-state performance to be degraded. Also, by plotting Eqs. (16), variations of $\mathcal{M}$ with respect to B_w and for different values of μ can be studied, as shown in Figure 2-b. From this figure, it can be seen that increasing B_w causes $\mathcal{M}$ to be decreased. For very narrow noise band-widths ($B_w<0.1$), $\mathcal{M}$ is highly decreased. However, for wider noise band-widths, the influence of B_w on $\mathcal{M}$ is less pronounced.

Also, it can be shown from Eq. (15) that $\hat{\mathbf{s}}=\mathbf{s}$ does not necessarily degrade the steady-state performance of FxLMS-based ANC systems. For improving the steady-state performance by $\hat{\mathbf{s}}$, $\mathcal{M}$ should be smaller than $\bar{\mathcal{M}}$. For small step-sizes, combining Eqs. (15) and (16) results in the following condition for having $\mathcal{M}<\bar{\mathcal{M}}$

$$\mathbf{s}^T\hat{\mathbf{s}}>\left\|\hat{\mathbf{s}}\right\|^2 \quad (17)$$

2. 2　Stability Analysis

For a stable adaptation process, the steady-state MSE should be greater than J_o. In this case, $J_{ex}(\infty)$ is always greater than zero. Therefore, by setting Eq. (13) to 0, an upper-bound for the step-size can be derived as

$$\mu_{max}=\frac{2\mathbf{s}^T\hat{\mathbf{s}}}{\left\|\mathbf{s}\right\|^2\left\|\hat{\mathbf{s}}\right\|^2\sigma_x^2\left(L+\frac{2}{B_w}\frac{\mathbf{s}^T\hat{\mathbf{s}}\mathbf{s}^T\mathbf{\Psi}\hat{\mathbf{s}}}{\left\|\hat{\mathbf{s}}\right\|^2\left\|\mathbf{s}\right\|^2}\right)} \quad (18)$$

For a perfect secondary path model, μ_{max} is simplified to

$\mu_{\bar{max}}$:

$$\mu_{\bar{max}} = \frac{2}{\|\mathbf{s}\|^2 \sigma_x^2 \left(L + \frac{2}{B_w} D_{eq}\right)} \tag{19}$$

As can be seen in Eq. (19), the influences of the secondary path on the stability bound is taken into account by $\|\mathbf{s}\|$ and D_{eq}. Usually, μ is normalized by $\|\mathbf{s}\|^2 \sigma_x^2$; therefore, D_{eq} and B_w can be considered as the major parameter of the secondary path which affects on μ_{max}. The variation of μ_{max} with respect to D_{eq} can be plotted by using Eq. (19), as shown in Figure 2-c. As can be seen in this figure, by increasing D_{eq}, μ_{max} is decreased.

Also, it can be seen from Eq. (19) that the proposed expression for μ_{max} is in the form of the commonly-used expression, derived by Elliott in [21]. In fact, Elliott derived the same expression, but instead of D_{eq}, the physical time delay in the secondary path appeared in his expression. This is because he assumed a pure delay secondary path but we consider an arbitrary secondary path in a general form.

Figure 2-c shows variations of μ_{max} with respect to D_{eq} for different values of B_w. This figure is obtained by plotting Eq. (18). The curve, shown by $B_w = 1$, corresponds to the case with a broad-band white noise. From this figure, it can be seen that by decreasing B_w, μ_{max} is also decreased. Also, by decreasing B_w, the influence of D_{eq} on μ_{max} is more pronounced.

According to Eq. (18), it is possible that the secondary path model causes μ_{max} to become negative. In this case, there is no positive step-size for which the FxLMS algorithm can become stable. Based on this logic, secondary path models causing the FxLMS algorithm to become unstable can be determined. Usually L is set to a large number, therefore it can be assumed that the denominator of Eq. (18) is always positive. Accordingly, the sign of μ_{max} can be determined only by considering the numerator of Eq. (18). Therefore, only for secondary path models with $\mathbf{s}^T\hat{\mathbf{s}} < 0$, the FxLMS algorithm cannot become stable. On the other hand, the inner product of $\mathbf{s}$ and $\hat{\mathbf{s}}$ can be expressed as

$$\mathbf{s}^T\hat{\mathbf{s}} = \|\mathbf{s}\| \, \|\hat{\mathbf{s}}\| \cos\phi \tag{20}$$

where ϕ is the angle between vectors $\mathbf{s}$ and $\hat{\mathbf{s}}$ in Q-dimensional space. Since both $\|\mathbf{s}\|$ and $\|\hat{\mathbf{s}}\|$ are positive, the instability condition of the FxLMS algorithm (which is $\mathbf{s}^T\hat{\mathbf{s}} < 0$) requires that

$$\cos\phi < 0 \tag{21}$$

As an elegant result, when the angle between the actual secondary path vector and its model vector is greater than 90°, then the FxLMS algorithm cannot become stable. This result is in an excellent agreement with the 90° condition, derived by Boucher for pure delay secondary paths and tonal acoustic noise [22] or by Morgan for an identity secondary path model [23]. For these simplified cases, in which only one element in vectors $\mathbf{s}$ and $\hat{\mathbf{s}}$ are non-zero, Boucher showed that if the secondary path model is 90° out of phase, then the FxLMS algorithm cannot become stable. However, in the derivation of Eq. (21), a general secondary path, its arbitrary model, and an arbitrary acoustic noise band-width are considered.

3. Experimental Results

Fig. 3 shows a general scheme for the implementation of single-channel feed-forward ANC. The microphone located close to the noise source is called the reference microphone and the one located in the desired zone of silence (Z_s) is called the error microphone. The outputs of these microphones are fed to A/D modules to produce $x(n)$ and $e(n)$. The A/D modules are connected to a FPGA chassis, where signal acquisition and conditioning algorithms can be performed. Also, the FPGA chassis produces $y(n)$ which is then fed to a D/A module. The output of the D/A module is then fed to a loudspeaker. The acoustic signal produced by the loudspeaker propagates through the medium to reach Z_s. Another hardware component of the proposed scheme is a real-time controller which is responsible for transferring data between the FPGA chassis and a monitoring PC station. The effects of the microphones, loudspeaker and other electro-acoustical components can be included in the primary and secondary paths [24]. In this case, the scheme, shown in Fig. 3, completely matches the model given in Fig. 1. We implemented this scheme in an acoustic duct, where the sound propagation can be modeled in one spatial dimension. Also, we use a BEQ acoustic head with a hearing system similar to the human one. This acoustic head does not contribute to ANC operations; but it is used for monitoring the ANC system performance. This efficient and flexible experimental setup can be used for the embedded implementation of different ANC algorithms. The error microphone is located very close to the right ear of the acoustic head and the signal heard by the acoustic head is recorded as a measure for evaluating the ANC system performance.

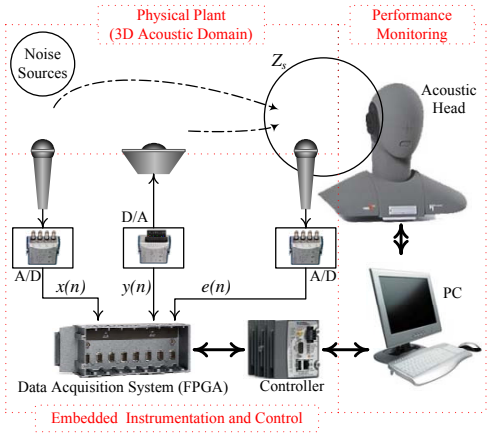


図 3　シングルチャネルフィードフォワード ANC の実装図

Fig. 3　A general scheme for embedded realization of single-channel feed-forward ANC

表 1　実 験 条 件

Table 1　Experimental conditions

	B_w	$\hat{\mathbf{s}}$	J_0
Case 1	0.36	perfect	0.0377 W
Case 2	0.18	perfect	0.0241 W
Case 3	0.18	model 1	0.0241 W
Case 4	0.18	model 2	0.0241 W

Using this experimental setup, several experiments in four different cases are conducted. In these experiments, the acoustic noise power is set to $\sigma_x^2 = 6$ W. The other experimental conditions for each case are detailed in Table 1. The two secondary path models mentioned in Table 1 are shown

表 2　ステップサイズの上限に関する実験結果と理論値の比較

Table 2　Comparison of experimental and theoretical values of step-size upper-bound

	Theory	Practice			
	μ_{max}	μ_M	error	μ_m	error
Case 1	0.001448	0.0013	−10.21%	0.00155	7.05%
Case 2	0.000979	0.00085	−13.16%	0.00105	−7.27%
Case 3	0.001360	0.0012	−11.78%	0.00145	−6.60%
Case 3	0.001901	0.0017	−10.58%	0.00205	−7.83%

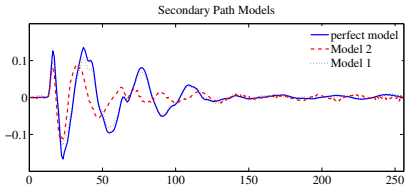


図 4　実験において用いた二次経路モデルのインパルス応答

Fig. 4　Secondary path models used in the experiments

in Fig. 4.

3. 1　System Stability in Practice

In the first case experiments, when $\mu < 0.0013$, the system is stable. Different experiments with different step-sizes show that when $0.0013 < \mu < 0.00155$ the system starts diverging after a short time (becomes unstable); and when $\mu > 0.00155$ the system starts diverging from the beginning. These observations are recorded in Table 2. According to this table, it can be concluded that a practical μ_{max} is located between 0.0013 and 0.00155. However, it is not technically possible to estimate any specific value for μ_{max} from the experimental results. This is because of the experimental conditions uncertainties such as changing characteristics of the surrounding environment, non-stationary behaviors of the background noise and uncertainties associated with physical plants, control systems and measurement devices.

Now, by substituting the parameters, given in Table 1, into Eq. (18), the theoretical value of μ_{max} can be obtained as $\mu_{max} = 0.001448$. It means that, in theory, for $\mu < 0.001448$ the system always becomes stable and for $\mu > 0.001448$, the algorithm always diverges. This result is in a good agreement with the experimental results described above. Here, the minimum relative error between the practical and theoretical results can be obtained as the relative difference of the theoretical value of μ_{max} and the maximum step-size for which the ANC system is stable in practice (e.g. this value is 0.0013 in the above experiment). Similarly, the maximum relative error can be obtained as the relative error between the theoretical value of μ_{max} and the minimum step-size for which the ANC system is always unstable in practice (e.g. this value is 0.00155 in the above experiment). The obtained values of μ_{max} and the corresponding relative errors are recorded in Table 2.

The above experimental process can be repeated when the ANC setup operates in the other three cases. The obtained results, along with the theoretical results are collected in Table 2. Also, the relative errors between the theoretical and experimental results are shown in this table. As can be seen in this table, the proposed formulation for μ_{max} can efficiently estimate this parameter with the relative error between −15% and 10%. The existence of this error can be justified because in theory it is not possible to consider uncertainties associated with physical plants, control systems and measurement mechanisms. Apart from this unavoidable error, the proposed formulation for μ_{max} is a very significant step towards matching available theoretical and experimental findings. This is because the commonly-used expression for μ_{max}, which was derived in [14], [21], can only apply to unrealistic cases with pure delay secondary paths, broad-band acoustic noise signals and perfect secondary path models. However, the proposed formulation can apply to a general case with an arbitrary secondary path, arbitrary acoustic noise band-width, and arbitrary secondary path model.

3. 2　Steady-State Performance in Practice

Initially, the implemented ANC system is activated in the case 1 experimental conditions and the step-size is set to a relatively small number (about 0.1 of its corresponding upper-bound). After a long time, when the system reaches its steady-state conditions, the power of the residual noise is computed in LabVIEW. The step-size is then increased incrementally and the system is re-started again. For each step-size, the above experiment is repeated and the steady-state power of the residual noise (MSE level) is measured. The measured data can be plotted as a function of the step-size, as shown in Figure 5a. For each step-size, the theoretical value of the steady-state residual acoustic noise can be also computed by using the closed-form expression given in Eq. (13) and the parameters given in Table 1. The results can be then plotted as another function of the step-size. This function is plotted by using a red line in Figure 5a. Also, the minimum attainable noise power (J_o) is shown in this figure .

For the other three cases, the same process can be repeated. The obtained experimental results are plotted in Figures 5b, 5c and 5d. As can be seen in plots of Figures 5, the theoretical and practical results are in a good agreement. The existence of the small offset between the experimental and theoretical results is due to the non-ideal experimental conditions. The agreement between the theoretical and experimental results, shown in Figure 5, confirms that the proposed formulations can effectively estimate the steady-state performance of practical FxLMS-based ANC systems.

4.　Conclusion

The prime goal of this study has been a comprehensive theoretical analysis of FxLMS-based ANC systems in realistic conditions in which actual ANC systems work. The attention has been focused on the derivation of general closed-form expressions for parameters relevant to stability behaviors of ANC systems, such as the step-size upper-bound and steady-state performance. These expressions have been derived based on a novel stochastic model for ANC systems. Relatively comprehensive expressions for the upper-bound of the step-size and the steady state performance have been derived. These expressions are in the form of the commonly-used ones, previously derived by Elliott for pure delay secondary paths. However, instead of the physical time delay which appeared in Elliott's expressions, a hypothetical parameter has appeared in the newly-derived expressions. This novel parameter, which is called the secondary path equivalent delay, can be computed for any arbitrary secondary path, unlike the secondary path time delay which is a physical parameter related to only pure delay secondary paths.

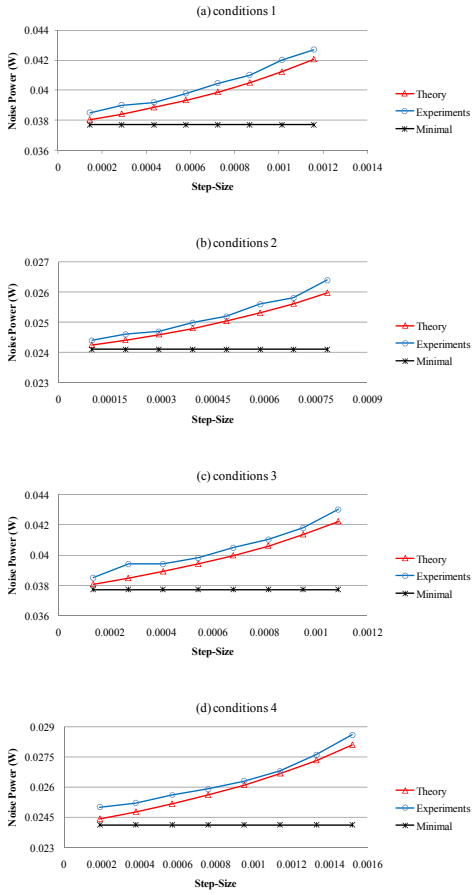


図 5　実験における定常状態での MSE

Fig. 5　Steady-state MSE in practice

The influences of the acoustic noise band-width and secondary path modeling error on the step-size upper-bound and the steady-state performance have been investigated. It has been shown that these influences can be included in the closed-form expressions, obtained.

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